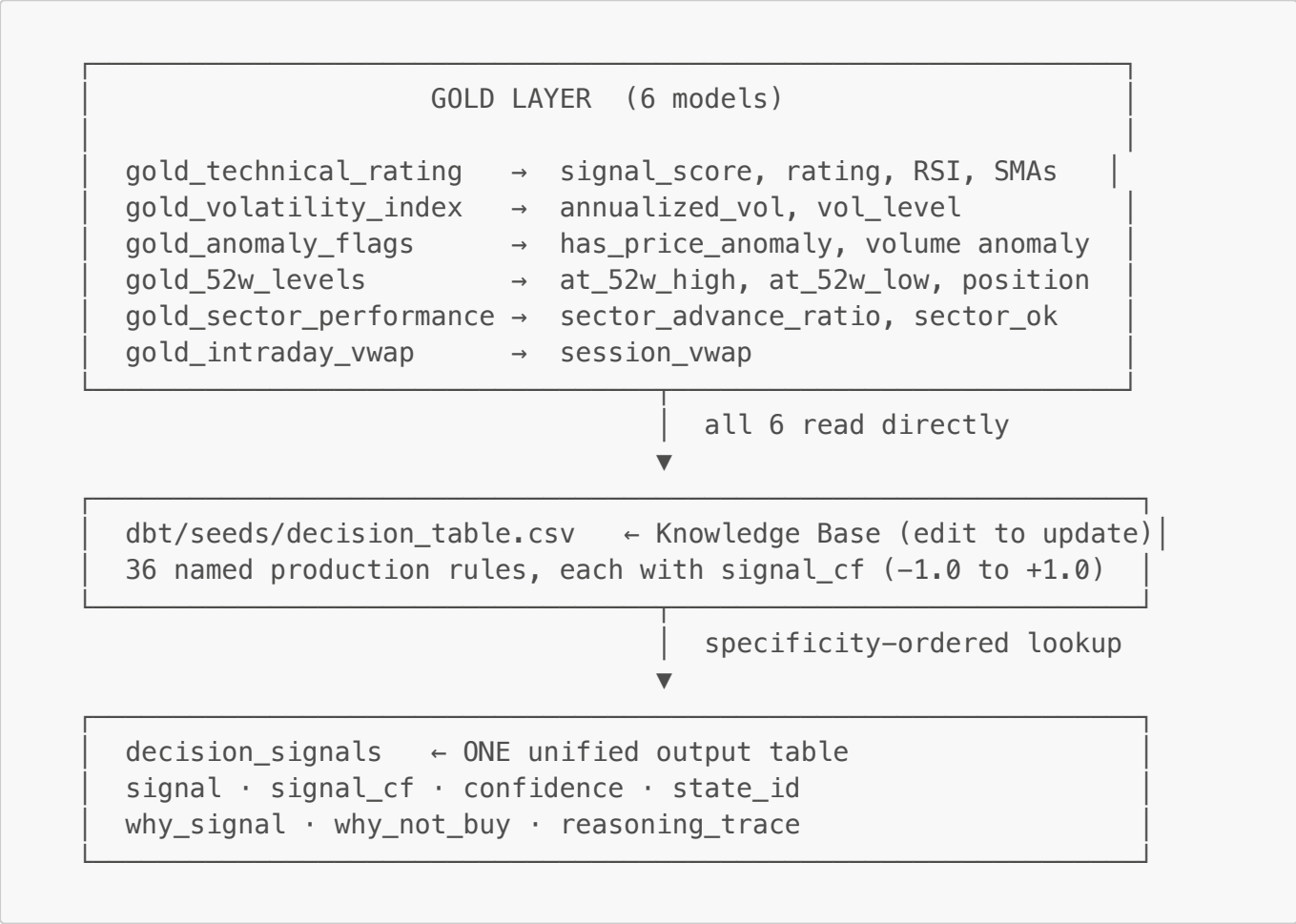


Decision Layer — Signal Methodology

This document defines the six Gold layer models that feed the decision layer, how their outputs are used as inputs to `dbt/seeds/decision_table.csv`, and how the single `decision_signals` model produces a **BUY / SELL / HOLD** signal with a formal certainty factor (−1.0 to +1.0) for every Tadawul symbol on every trading day.

Architecture



Knowledge representation used (Lecture 4):

Technique	Where	§
Decision table	<code>decision_table.csv</code> — 36 named IF-THEN rules with CF	§4.7 Formal Logic
Decision tree	<code>signal_score</code> → <code>rating</code> — 5-branch CASE in SQL	§4.7 Formal Logic
Production rules	Each row of the decision table is a production rule	§4.1
Certainty Factors	<code>signal_cf</code> column — formal uncertainty (−1.0 to +1.0)	§4.10

Part 1 — The Six Gold Layer Models

1. gold_technical_rating

Computes **8 indicator votes** (+1 buy, 0 neutral, -1 sell) from the daily close price and aggregates them into a composite score and rating label.

Indicators and vote conditions:

Rule	Indicator	Buy vote (+1)	Sell vote (-1)
R01	Price vs SMA10	close > SMA10	close < SMA10
R02	Price vs SMA20	close > SMA20	close < SMA20
R03	Price vs SMA50	close > SMA50	close < SMA50
R04	Price vs SMA200	close > SMA200	close < SMA200
R05	MA cross	SMA10 > SMA20 (golden cross)	SMA10 < SMA20 (death cross)
R06	RSI(14)	RSI < 30 — oversold	RSI > 70 — overbought
R07	Bollinger Bands	close < BB_lower	close > BB_upper
R08	MACD proxy	SMA12 > SMA26	SMA12 < SMA26

Score and rating:

```
signal_score = Σ votes    (range: -8 to +8)

≥ +5  → Strong Buy
≥ +3  → Buy
≤ -5  → Strong Sell
≤ -3  → Sell
else  → Neutral
```

Key columns used in decision_signals:

Column	Type	Used for
signal_score	INT	Decision tree classification → rating
buy_signals	INT	Explanation output
sell_signals	INT	Explanation output + anomaly SELL override
rsi14	DOUBLE	Explanation output
sma50, sma200	DOUBLE	Explanation output
bb_lower, bb_upper	DOUBLE	Explanation output

2. gold_volatility_index

Computes **rolling volatility** from daily log-returns.

$$\sigma_{r_t} = \ln\left(\frac{\text{close}_t}{\text{close}_{t-1}}\right) \quad \text{annualized_vol} = \sigma_{20d} \times \sqrt{252}$$

Key columns used in **decision_signals**:

Column	Type	Used for
annualized_vol	DOUBLE	Classified into vol_level — decision table key
log_return	DOUBLE	Anomaly SELL override (score check)

vol_level classification (decision table key):

annualized_vol > 80%	→	extreme	(daily swings > 5% – unsafe entry)
annualized_vol > 50%	→	high	(elevated – reduces buy conviction)
annualized_vol > 20%	→	normal	(standard trading environment)
annualized_vol ≤ 20%	→	low	(calm – highest buy confidence)

3. **gold_anomaly_flags**

Detects **abnormal price and volume behaviour** using two SQL-layer methods.

Volume Z-score:

$$z = \frac{V_t - \bar{V}_{30d}}{\sigma_{V,30d}} \quad \text{flags when } |z| > 2.5$$

Price IQR (Tukey fence on 90-day trailing log-returns):

$$\text{lower} = Q_1 - 1.5(Q_3 - Q_1) \quad \text{upper} = Q_3 + 1.5(Q_3 - Q_1)$$

Flags when daily log-return falls outside **[lower, upper]**.

Key columns used in **decision_signals**:

Column	Type	Used for
has_price_anomaly	BOOLEAN	Decision table key (has_anomaly) · anomaly SELL override
has_volume_anomaly	BOOLEAN	Explanation output

4. **gold_52w_levels**

Computes **rolling 52-week high and low** over a 252-trading-day window.

$$\text{high}_{52w} = \max(\text{high}_{t-251}, \dots, \text{high}_t) \quad \text{low}_{52w} = \min(\text{low}_{t-251}, \dots, \text{low}_t)$$

$$\text{pct_from_high} = \frac{\text{close} - \text{high}_{52w}}{\text{high}_{52w}} \quad \text{pct_from_low} = \frac{\text{close} - \text{low}_{52w}}{\text{low}_{52w}}$$

Proximity flags within 2%: **at_52w_high**, **at_52w_low**.

Key columns used in `decision_signals`:

Column	Type	Used for
<code>at_52w_high</code>	BOOLEAN	Classified into <code>at_52w_pos</code> — decision table key
<code>at_52w_low</code>	BOOLEAN	Classified into <code>at_52w_pos</code> — decision table key
<code>high_52w, low_52w</code>	DOUBLE	Explanation output
<code>pct_from_high, pct_from_low</code>	DOUBLE	Explanation output

`at_52w_pos` classification (decision table key):

```

at_52w_low = true  → near_low  (within 2% of 52W low – mean reversion
zone)
at_52w_high = true → near_high (within 2% of 52W high – resistance zone)
otherwise         → neutral   (between extremes)

```

5. `gold_sector_performance`

Computes **sector-wide momentum** across all stocks in each Tadawul sector.

$$\text{advance_ratio} = \frac{\text{\# stocks with positive daily return}}{\text{\# total stocks in sector}}$$

A value ≥ 0.5 means the sector is broadly advancing.

Key columns used in `decision_signals`:

Column	Type	Used for
<code>advance_ratio</code>	DOUBLE	Classified into <code>sector_ok</code> ($\geq 0.5 = \text{true}$) — decision table key

6. `gold_intraday_vwap`

Computes **session Volume-Weighted Average Price** from tick data.

$$\text{VWAP} = \frac{\sum_t \text{price}_t \times \text{volume}_t}{\sum_t \text{volume}_t}$$

Key columns used in `decision_signals`:

Column	Type	Used for
<code>session_vwap</code>	DOUBLE	Passed through as context column — not a decision table key

VWAP provides intraday context. It is available only during market hours (ticks from the Kafka producer). **NULL** outside trading sessions.

Part 2 — Decision Table Lookup

The Knowledge Base: **decision_table.csv**

Each row is a **named production rule** of the form:

```
IF  rating_group = X
AND has_anomaly  = Y      (true / false / any)
AND sector_ok    = Z      (true / false / any)
AND vol_level    = V      (low / normal / high / extreme / any)
AND at_52w_pos   = P      (near_low / neutral / near_high / any)
THEN signal = S,  signal_cf = CF
```

The **signal_cf** value IS the formal certainty factor (§4.10) — a numeric scale from −1.0 to +1.0 pre-assigned based on domain expertise:

```
+1.0  maximum BUY conviction
|
+0.65  HIGH confidence threshold
|
+0.40  MEDIUM confidence threshold
|
0.00   no directional conviction → HOLD
|
-0.40  MEDIUM confidence threshold
|
-0.65  HIGH confidence threshold
|
-1.0   maximum SELL conviction
```

Structure: 7 rows per rating group — symmetric coverage

Each of the 5 rating groups gets **7 rows** covering the same 7 scenarios:

Row within group	Scenario	state_id range
1	High volatility gate (blocks or modifies)	S01, S08, S15, S22, S29
2	Extreme volatility gate (stronger block)	S02, S09, S16, S23, S30
3	Price anomaly gate	S03, S10, S17, S24, S31
4	Sector headwind gate	S04, S11, S18, S25, S32
5	Near 52W low differentiator (mean reversion)	S05, S12, S19, S26, S33
6	Near 52W high differentiator (resistance/breakdown)	S06, S13, S20, S27, S34
7	Default for this rating group	S07, S14, S21, S28, S35

Plus **S36** — the global fallback row (**any** for all conditions, HOLD, CF=0.00).

Total: 36 rows

The 36 Rules

Strong Buy group (S01–S07):

State	has_anomaly	sector_ok	vol	at_52w	Signal	CF	Conf
S01	false	true	high	any	HOLD	+0.22	LOW
S02	false	true	extreme	any	HOLD	−0.12	LOW
S03	true	any	any	any	HOLD	−0.20	LOW
S04	false	false	any	any	HOLD	+0.12	LOW
S05	false	true	any	near_low	BUY	+0.92	HIGH
S06	false	true	any	near_high	BUY	+0.72	HIGH
S07	false	true	any	any	BUY	+0.88	HIGH

Buy group (S08–S14):

State	has_anomaly	sector_ok	vol	at_52w	Signal	CF	Conf
S08	false	true	high	any	HOLD	+0.18	LOW
S09	false	true	extreme	any	HOLD	−0.12	LOW
S10	true	any	any	any	HOLD	−0.28	LOW
S11	false	false	any	any	HOLD	0.00	LOW
S12	false	true	any	near_low	BUY	+0.80	HIGH
S13	false	true	any	near_high	BUY	+0.55	MEDIUM
S14	false	true	any	any	BUY	+0.70	HIGH

Neutral group (S15–S21):

State	has_anomaly	sector_ok	vol	at_52w	Signal	CF	Conf
S15	false	true	high	any	HOLD	−0.10	LOW
S16	false	true	extreme	any	HOLD	−0.18	LOW
S17	true	any	any	any	HOLD	−0.20	LOW
S18	false	false	any	any	HOLD	−0.08	LOW
S19	false	true	any	near_low	HOLD	+0.12	LOW
S20	false	true	any	near_high	HOLD	−0.08	LOW
S21	any	any	any	any	HOLD	0.00	LOW

Sell group (S22–S28):

State	has_anomaly	sector_ok	vol	at_52w	Signal	CF	Conf
S22	false	true	high	any	SELL	−0.68	HIGH
S23	false	true	extreme	any	SELL	−0.60	MEDIUM
S24	true	any	any	any	SELL	−0.85	HIGH
S25	false	false	any	any	SELL	−0.75	HIGH
S26	false	true	any	near_low	SELL	−0.52	MEDIUM
S27	false	true	any	near_high	SELL	−0.82	HIGH
S28	any	any	any	any	SELL	−0.70	HIGH

Strong Sell group (S29–S35):

State	has_anomaly	sector_ok	vol	at_52w	Signal	CF	Conf
S29	false	true	high	any	SELL	−0.80	HIGH
S30	false	true	extreme	any	SELL	−0.72	HIGH
S31	true	any	any	any	SELL	−0.95	HIGH
S32	false	false	any	any	SELL	−0.88	HIGH
S33	false	true	any	near_low	SELL	−0.68	HIGH
S34	false	true	any	near_high	SELL	−0.92	HIGH
S35	any	any	any	any	SELL	−0.90	HIGH

Fallback:

State	All conditions	Signal	CF	Conf
S36	any / any / any / any / any	HOLD	0.00	LOW

Row Matching: Specificity-Ordered Lookup

Each (symbol, date) is joined against all matching rows. The row with the highest **specificity_score** wins. When two rows tie, the **lower state_id** wins.

```
specificity_score = count of non-'any' conditions
                  = (rating_group ≠ 'any') + (has_anomaly ≠ 'any')
                  + (sector_ok ≠ 'any') + (vol_level ≠ 'any')
                  + (at_52w_pos ≠ 'any')
```

Designed tie-breaking: Within each group, gate rows (S_01–S_04 etc.) have lower state_ids than differentiator rows (S_05–S_07 etc.). Both have specificity=4 for the vol vs 52W case. The lower state_id (vol gate) wins — meaning **high volatility always blocks BUY even when near 52W low**.

Example resolutions:

Strong Buy + no anomaly + sector_ok + high vol + near_low:
S01 (high vol, spec=4) vs S05 (near_low, spec=4) → S01 wins → HOLD,
CF=+0.22

Strong Buy + no anomaly + sector_ok + normal vol + near_low:
S01 doesn't match (vol≠high) → S05 wins → BUY, CF=+0.92

Strong Buy + anomaly + near_low:
S03 (anomaly=true, spec=2) vs S05 (anomaly=false → doesn't match) → S03
wins → HOLD, CF=-0.20

Strong Buy + no anomaly + sector_fail + near_low:
S04 (sector_ok=false, spec=3) vs S05 (sector_ok=true → doesn't match) →
S04 wins → HOLD, CF=+0.12

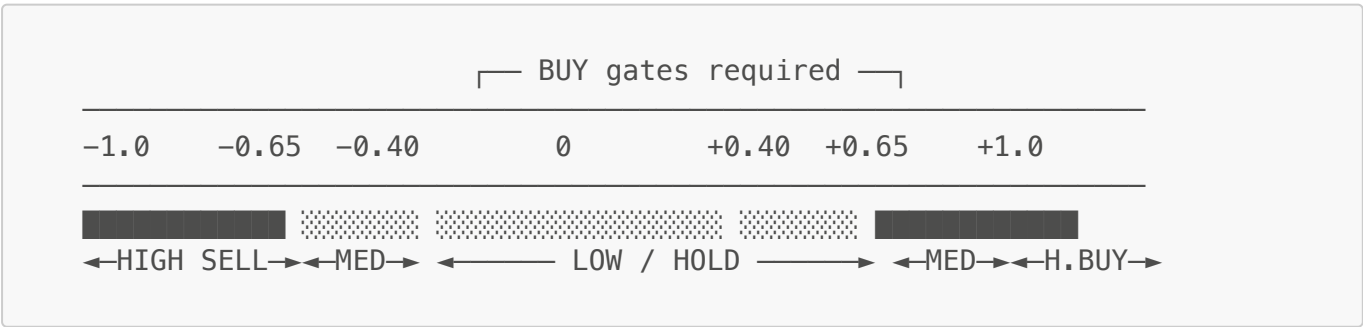
Part 3 — Anomaly SELL Override

After the table lookup, one additional production rule fires before the final signal is set. It handles a scenario not in the decision table: a **Neutral-rated stock** (score in range -2 to 0) with a simultaneous price-IQR anomaly:

IF has_price_anomaly = TRUE
AND signal_score < 0
THEN signal = SELL, signal_cf = -0.50, confidence = MEDIUM

This covers states like S17 (Neutral + anomaly → HOLD, CF=-0.20) where the table conservatively gives HOLD, but the combination of anomaly AND a negative score constitutes a SELL trigger per the original system logic.

Part 4 — CF Number Line



signal_cf	confidence	Example states
≥ 0.65	HIGH	S05 (+0.92), S31 (-0.95), S34 (-0.92), S07 (+0.88)

signal_cf	confidence	Example states
≥ 0.40	MEDIUM	S13 (+0.55), S23 (−0.60), S26 (−0.52)
< 0.40	LOW	S01 (+0.22), S04 (+0.12), S17 (−0.20), S21 (0.00)

Part 5 — Explanation Facility (§4.8–§4.9)

why_signal — Dynamic (reconstructed from matched row)

```
"BUY (CF=0.92, HIGH): state=S05 — Best setup: strong technicals + clean
conditions + near 52W low"
"SELL (CF=−0.85, HIGH): state=S24 — Sell rating + price anomaly =
compounded bearish confirmation"
"HOLD (CF=0.22, LOW): state=S01 — High volatility blocks BUY — position
sizing unsafe"
```

why_not_buy — First blocking condition (NULL when signal = BUY)

```
"Rating is 'Neutral' (score=1/8) — not eligible for BUY"
"Price-IQR anomaly detected — spike entry risk (state S10)"
"Sector advancing 43% < 50% threshold"
"high volatility (62.4% ann.) — position risk"
"extreme volatility (83.1% ann.) — position risk"
```

reasoning_trace — Numbered How trace (§4.9)

```
1. Inputs classified: rating=Strong Buy, score=+6/8, vol=normal,
   52W=near_low, anomaly=false, sector_ok=true.
2. Decision table matched: state=S05 (specificity=4).
3. Signal=BUY, CF=0.92 (HIGH).
```

```
1. Inputs classified: rating=Buy, score=+3/8, vol=high,
   52W=neutral, anomaly=false, sector_ok=true.
2. Decision table matched: state=S08 (specificity=4).
3. Signal=HOLD, CF=0.18 (LOW). [high vol gate fired]
```

Output Columns

Column	Type	Description
symbol	VARCHAR	4-digit Tadawul code

Column	Type	Description
company_name	VARCHAR	Full company name
sector	VARCHAR	Sector classification
date	DATE	Trading date
close	DOUBLE	Closing price (SAR)
signal	VARCHAR	BUY / SELL / HOLD
signal_cf	DOUBLE	Certainty factor −1.0 to +1.0 (§4.10)
confidence	VARCHAR	HIGH (≥0.65) / MEDIUM (≥0.40) / LOW (<0.40)
state_id	VARCHAR	Matched decision table row (e.g. S05)
why_signal	VARCHAR	Dynamic explanation of this signal
why_not_buy	VARCHAR	What blocked BUY — NULL if signal = BUY
why_not_sell	VARCHAR	Why SELL not triggered — NULL if signal = SELL
reasoning_trace	VARCHAR	Numbered step-by-step inference trace
rating	VARCHAR	Strong Buy / Buy / Neutral / Sell / Strong Sell
signal_score	INT	−8 to +8 net indicator vote
buy_signals	INT	Count of BUY votes (0–8)
sell_signals	INT	Count of SELL votes (0–8)
rsi14	DOUBLE	14-day RSI
sma50 / sma200	DOUBLE	Simple moving averages
bb_lower / bb_upper	DOUBLE	Bollinger Band boundaries
annualized_vol	DOUBLE	20-day vol × $\sqrt{252}$
vol_level	VARCHAR	low / normal / high / extreme
high_52w / low_52w	DOUBLE	52-week extremes
pct_from_high / pct_from_low	DOUBLE	% distance from 52W extremes
at_52w_high / at_52w_low	BOOLEAN	Within 2% of 52W extreme
at_52w_pos	VARCHAR	near_low / neutral / near_high
has_price_anomaly	BOOLEAN	IQR outlier on daily return
has_volume_anomaly	BOOLEAN	Z-score > 2.5 on volume
sector_advance_ratio	DOUBLE	Fraction of sector advancing (0–1)
sector_ok	BOOLEAN	sector_advance_ratio ≥ 0.50
session_vwap	DOUBLE	Intraday VWAP from tick stream

Query Examples

```
-- Today's BUY signals ranked by CF
SELECT symbol, company_name, sector, close, signal_cf, state_id, why_signal
FROM iceberg.decision.decision_signals
WHERE date = (SELECT MAX(date) FROM iceberg.decision.decision_signals)
  AND signal = 'BUY'
ORDER BY signal_cf DESC;

-- Which decision table row matched each symbol today?
SELECT symbol, rating, vol_level, at_52w_pos,
       has_price_anomaly, sector_ok, state_id, signal, signal_cf
FROM iceberg.decision.decision_signals
WHERE date = (SELECT MAX(date) FROM iceberg.decision.decision_signals)
ORDER BY signal_cf DESC;

-- Buy/Strong Buy stocks that got blocked – what state matched?
SELECT symbol, rating, signal_score, vol_level, at_52w_pos,
       has_price_anomaly, sector_ok, state_id, signal_cf, why_not_buy
FROM iceberg.decision.decision_signals
WHERE date = (SELECT MAX(date) FROM iceberg.decision.decision_signals)
  AND signal = 'HOLD'
  AND rating IN ('Buy', 'Strong Buy')
ORDER BY signal_cf DESC;

-- Distribution of matched states today
SELECT state_id, signal, confidence, COUNT(*) AS symbols_matched
FROM iceberg.decision.decision_signals
WHERE date = (SELECT MAX(date) FROM iceberg.decision.decision_signals)
GROUP BY state_id, signal, confidence
ORDER BY symbols_matched DESC;

-- Full trace for a specific symbol
SELECT date, signal, state_id, signal_cf, confidence, reasoning_trace
FROM iceberg.decision.decision_signals
WHERE symbol = '2222'
ORDER BY date DESC LIMIT 10;
```

Updating the Knowledge Base

To adjust a signal, change a CF value, or add a new rule:

1. **Edit** `dbt/seeds/decision_table.csv`
2. **Run:**

```
docker exec dbt dbt seed --select decision_table --project-dir /usr/dbt --
profiles-dir /root/.dbt
```

```
docker exec dbt dbt run --select decision_signals --project-dir /usr/dbt
--profiles-dir /root/.dbt
```

No SQL knowledge required. The domain expert edits the CSV only.

CF assignment guide:

Scenario	Recommended signal_cf
Perfect BUY: strong rating + low vol + near 52W low	+0.90 to +0.95
Standard BUY: good conditions, neutral 52W	+0.70 to +0.88
Moderate BUY: near resistance or slightly elevated vol	+0.50 to +0.65
BUY blocked by gate — positive lean	+0.10 to +0.28
BUY blocked by gate — neutral	0.00
BUY blocked by anomaly — negative lean	−0.12 to −0.30
Standard SELL	−0.68 to −0.75
SELL near support (near 52W low) — lower conviction	−0.50 to −0.60
SELL confirmed by anomaly or sector	−0.80 to −0.88
Maximum SELL: Strong Sell + anomaly	−0.90 to −0.95

Adding a new rule — add a row with a new **state_id** (e.g. **S37**) and assign **specificity_score** as the count of non-**any** conditions. State_ids S01–S36 are reserved; new rows should use S37+ to avoid changing existing tiebreaker ordering.

Limitations

- **Not financial advice.** Signals are deterministic SQL rules on historical prices.
- **Shallow knowledge (§6.6).** No causal model — earnings surprises, news, and geopolitical events are not modelled.
- **Simulation data.** The Kafka producer uses random-walk simulation outside Tadawul market hours (Sun–Thu 10:00–15:00 Riyadh time). Signals from simulated ticks are for pipeline testing only.
- **VWAP gaps.** **session_vwap** is NULL for days with no tick data (batch-only dates).
- **CBR requires history.** **decision_validation** is empty until the Airflow CBR DAG has accumulated 30+ days of WIN/LOSS outcomes.
- **No look-ahead bias.** All indicators use only information available at market close on the signal date.